

Product Carbon Footprint Study Report

Product Name: power saw production, without catalytic converter

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: power saw production, without catalytic converter

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 Item(s) power saw, without catalytic converter

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 Item(s) power saw, without catalytic converter.

2. System Boundary

power saw production, without catalytic converter, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of power saw production, without catalytic converter produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **170.741 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

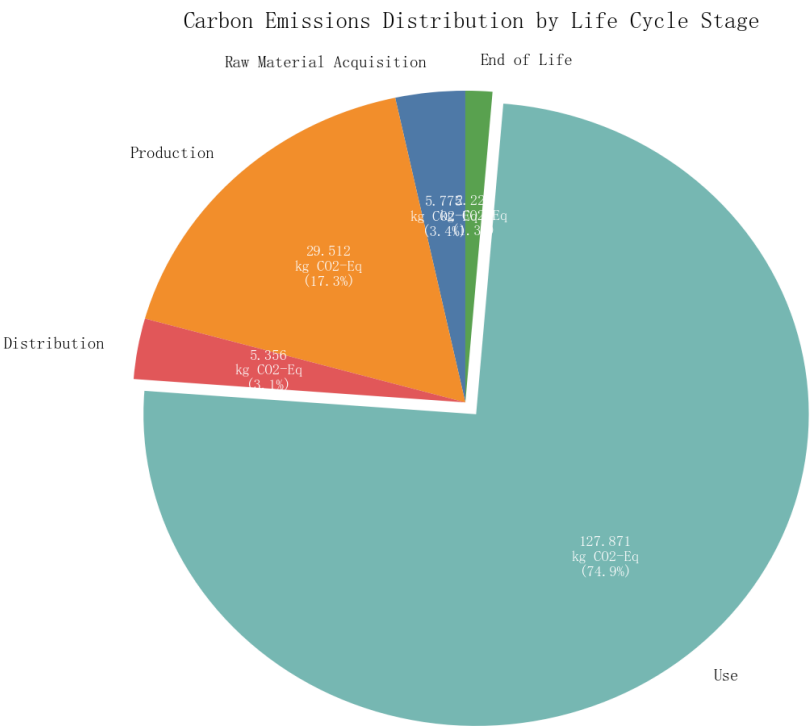


Table 1: Carbon Emissions by Life Cycle Stage

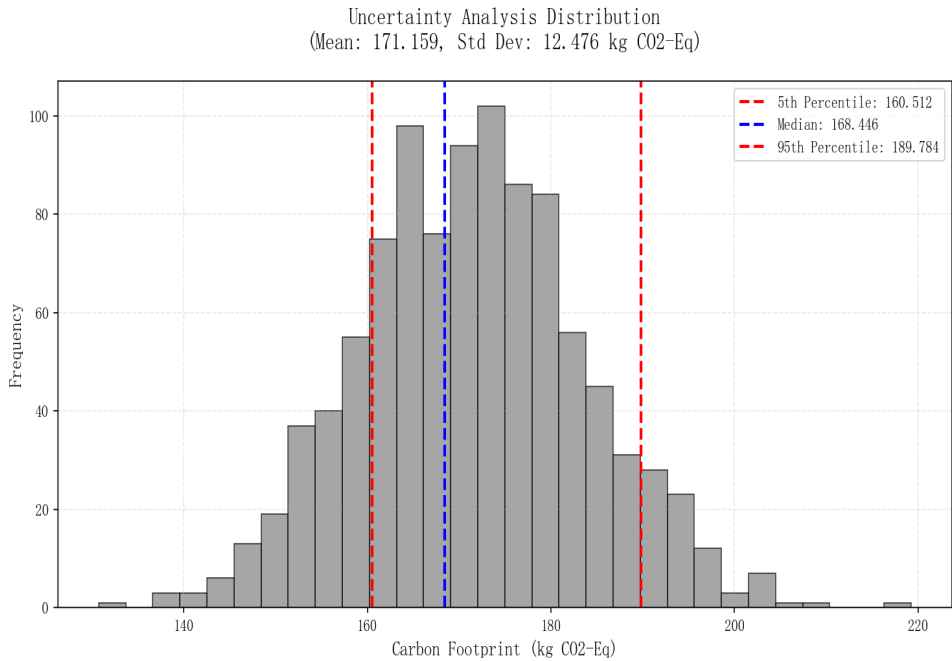
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	5.775	3.38
Production	29.512	17.28
Distribution	5.356	3.14

Use	127.871	74.89
End of Life	2.227	1.30
Total	170.741	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	171.159
Standard Deviation	12.476
Coefficient of Variation (CV)	7.29%
Median	168.446
5th Percentile	160.512
95th Percentile	189.784

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the power saw production, without catalytic converter, is calculated at 170.741 kg CO₂-Eq. This represents a moderate environmental impact for a power tool of this type, likely driven by energy consumption during the use phase. The lifecycle stage contributions reveal that the use phase is overwhelmingly dominant, accounting for 127.871 kg CO₂-Eq (74.9%) of total emissions. This suggests that the product's environmental performance is highly dependent on operational energy efficiency and user behavior, rather than embedded impacts from materials or manufacturing. Production contributes 29.512 kg CO₂-Eq (17.3%), while raw material extraction (5.775 kg CO₂-Eq, 3.4%) and distribution (5.356 kg CO₂-Eq, 3.1%) have relatively minor roles. The end-of-life stage is negligible at 2.227 kg CO₂-Eq (1.3%). Major emission sources are predominantly tied to the use phase, likely from electricity or fuel consumption during operation, with secondary contributions from manufacturing processes such as assembly and component fabrication. Given the lack of a catalytic converter, combustion-related emissions during use may be higher, emphasizing the need for efficiency improvements.

2. Statement of Assumptions and Limitations

Key assumptions in this study include that the use phase duration and intensity are based on typical consumer usage patterns for power saws, such as average operation hours per year and assumed fuel or electricity mix. The model assumes no catalytic converter is used, which may increase direct emissions from fuel combustion if the saw is gasoline-powered. Limitations arise from data quality: the top process contributions are listed as "Unknown" with zero values, indicating a gap in detailed process-level data, potentially from modeling shortcuts or lack of supplier-specific information. This reduces certainty in identifying exact emission hotspots within manufacturing or material supply chains. The 90% confidence interval (160.512–189.784 kg CO₂-Eq) and coefficient of variation (7.29%) show moderate uncertainty, likely due to variability in use phase assumptions. System boundary restrictions include a cradle-to-grave scope from raw material extraction to end-of-life, but exclude capital goods, infrastructure, and potential recycling credits. Additionally, the absence of a catalytic converter implies that emissions from combustion byproducts (e.g., particulates) are not fully captured, focusing only on CO₂-Eq.

3. Suggestions for Improvement

- ****Improve use phase efficiency****: Focus on reducing emissions during operation by designing the power saw with a higher-efficiency engine (if gasoline-powered) or optimizing motor and battery systems (if electric). Consider integrating a catalytic converter despite the current configuration to lower combustion byproducts.
- ****Optimize production processes****: Target the production stage (17.3% of emissions) by shifting to renewable energy sources for manufacturing facilities, implementing lean manufacturing to reduce waste, and selecting lower-carbon materials for components like housings or blades.
- ****Enhance data quality****: Resolve the "Unknown" process contributions by collecting detailed supplier data or conducting a process-level attributional LCA. This will enable precise identification of emission reduction opportunities in raw material supply and production.

- ****Extend product lifespan****: Encourage repairability and modular design to delay replacement, indirectly reducing lifecycle emissions per functional unit by spreading manufacturing impacts over longer use periods.